b. phellogen
**c. intercalary meristem**
d. intrafascicular cambium

15. A common feature of vessel elements and sieve tube elements is (CBSE- AIPMT 2007)
**a. enucleate condition**
b. presence of P. Protein
c. thick secondary wall
d. pores on lateral walls

16. In a longitudinal section of a root, starting from the tip upward, the four zones occur in the following order (CBSE -AIPMT 2004)
**a. root cap, cell division, cell enlargement, cell maturation**
b. root cap, cell division, cell maturation, cell enlargement
c. cell division, cell enlargement, cell maturation, root cap
d. cell division, cell maturation, cell enlargement, root cap

17. The cells of the quiescent centre are characterized by (CBSE -AIPMT 2003)
**a. having dense cytoplasm and prominent nucleus**
b. having light cytoplasm and small nucleus
c. dividing regularly to add to the corpus
d. dividing regularly to add to tunica

18. P. Protein is found in (CBSE- AIPMT 2000)
a. parenchyma b. collenchyma
**c. sieve tube** d. xylem

19. Specialized epidermal cells surrounding the guard cells are called (NEET (I) 2016)
a. bulliform cells
b. lenticels
c. complementary cells
**d. subsidiary cells**

**Directions:**
The following questions 20 & 21 consist of two statements, one labelled **Assertion** and the another labelled **Reason**. Select the correct answer from the codes given below:

a) Both assertion and reason are true and reason is the correct explanation of assertion
b) Both assertion and reason are true, but reason is not the correct explanation of assertion
c) Assertion is true but reason is false
d) Assertion and reason are false

20. **Assertion:** Conducting tissues, especially xylem show greatest reduction in submerged hydrophytes.
**Reason:** Hydrophytes live in water. So no need of tissues. (AIIMS – 2010) Ans: c.

21. **Assertion:** Long distance flow of photo assimilates in plants occurs through sieve tubes.
**Reason:** Mature sieve tubes have partial cytoplasm and perforated sieve plates (AIIMS – 2012)
Ans: a.

22. Duramen is present in (JIPMER 2016)
**a. the inner region of secondary wood**
b. a part of sap wood
c. the outer region of secondary wood
d. region of pericycle

23. The interxylary phloem is found in the stem of (JIPMER 2013)
a. Cucurbita b. Salvia
**c. Calotropis** d. none of these

24. Wound healing is due to (JIPMER 2013)
a. ventral meristem
**b. secondary meristem**
c. primary meristem
d. all of these

25. Which of the following tissues consists of living cells (JIPMER 2012)
a. vessels b. tracheids
**c. companion cell** d. sclerenchyma

26. The Quiescent centre in root meristem serves as a (JIPMER 2011)
a. site for storage of food, which is utilized during maturation
b. reservoir of growth hormones
**c. reserve for replenishment of damaged cells of the meristem**
d. region for absorption of water

27. In the sieve elements, which one of the following is the most likely function of P.Proteins? (JIPMER 2011)
a) Deposition of callose on sieve plates
b. Providing energy for active translocation
c. Autolytic enzymes
**d. Sealing-off mechanism on wounding**

28 .Which of the following is made up of dead cells? (NEET 2017)
a. Xylem parenchyma b. Collenchyma
**c. Phellem** d. Phloem

29. The vascular cambium normally gives rise to (NEET 2017)
a. phelloderm b.primary phloem
**c. secondary xylem** d. periderm

30. Which of the following plants shows multiple epidermis? (Manipal 2012)
a. Croton b. Allium **c. Nerium** d. Cucurbita

## UNIT V – Plant Physiology

1. The water potential of pure water is (NEET 2017)
a. Less than zero
b. More than zero but less than one
c. More than one **d. Zero**

2. Transpiration and root pressure cause water to rise in plants by (NEET 2015)
a. pulling it upward
**b. pulling and pushing it, respectively**



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